

Tao Financial Engine (TFE) Specification v2.6

1. Document Control

Document Title: Tao Financial Engine (TFE) Specification

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System: Tao Financial Engine (TFE)

Purpose: Normative specification for TFE as a research instantiation of DSF-AI

Scope: Current production profile (CP-0), transitional profile (CP-1), and thesis-faithful target profile (CP-2)

1.1 Intended Use

TFE is a research and governed advisory-support system that interprets structural geometry derived from financial time-series and associated governed runtime data. TFE is not a fully autonomous trading system. It is intended to support recommendation generation, portfolio guidance, runtime quality evaluation, and controlled policy learning under explicit operational gates.

1.2 Compliance Orientation

This specification is written to support: - deterministic system definition - good documentation practice - data integrity and auditability - change control and traceability - controlled deployment and validation

It is not, by itself, a claim of regulatory approval or production suitability for retail advisory use without further controls.

2. Executive Summary

TFE is built around a canonical structural engine (L0-L4) and a governed policy/runtime layer (L5). L0-L4 produce deterministic structural state from financial time-series. L5 consumes structural outputs and runtime context to produce governed decisions using explicit matching, fallback, validation, and publication rules.

The current production system is not yet the full thesis-faithful event-tape/shared-state DSF-AI target. It is a governed production profile that uses structural cell mapping, explicit candidate key chains, typed fallback semantics, promotion gates, runtime publication artifacts, and advisor surfaces. This specification therefore distinguishes between:

- **CP-0:** Current production code-truth profile
- **CP-1:** Transitional hybrid profile
- **CP-2:** Thesis-faithful target profile

The current competitiveness ceiling is not explained by provenance ambiguity or exact-path drift anymore. Those have been addressed. The main remaining ceilings are: - rulebook breadth - structural recency not

yet materialized into live runtime rows - epoch transport/join not yet realized - quote publication alignment/freshness operational hardening - incomplete public proof/quality planes

3. Definitions and Acronyms

3.1 Core Terms

DSF-AI — Deterministic Structural Field AI.

UF — Canonical structural engine producing structural states.

PSCF — Policy structural cell framework used for policy keying and runtime matching.

SES — Protected boundary and redacted summary discipline for kernel outputs.

Publication — Atomic runtime bundle containing aligned artifacts used by readers.

Exact Match — Runtime decision matched through exact structural family anchor.

Relaxed Match — Runtime decision matched through allowed relaxation path.

Fallback — Safe path used when preferred structural match is unavailable.

Epoch — Governed macro/event/sphere-of-impact state projected into runtime context.

Structural Recency — Deterministic counters describing elapsed structural time since key events.

3.2 Ladder Terms

- **L0_EXACT_STRUCTURAL_MATCH**
- **L1_RELAXED_STRUCTURAL_MATCH**
- **L2_SAFE_POLICY_FALLBACK**
- **L3_DEGRADED_UNMAPPED**
- **L4_UNAVAILABLE_OR_BLOCKED**

4. System Purpose and Architectural Philosophy

TFE exists to evaluate financial instruments using structural geometry rather than purely statistical prediction. Its governing principles are:

1. deterministic structural interpretation is primary
2. runtime advice must be bounded by explicit guards and fallbacks
3. freshness, publication alignment, provenance, and validation are first-class runtime obligations
4. reads should serve from a stable validated publication rather than recomputing expensive logic across inconsistent sources
5. current production truth must be distinguished from thesis-faithful target architecture

5. Conformance Profiles

5.1 CP-0 Current Production Profile

Current production L5 is a governed structural policy runtime with: - structural cell keying from serialized structural state - replay/row-trace policy generation options - explicit guards for action selection - promotion by non-regression gates - typed fallback ladder - runtime sync into Postgres-backed advisor surfaces - recommendations and portfolio surfaces with explicit degraded/fallback semantics

5.2 CP-1 Transitional Hybrid Profile

CP-1 includes experimental and hybridized temporal structures: - temporal datasets - row-trace-derived policy paths - hybrid replay/rowtrace learning modes - conformance microprobes - partial recency/epoch scaffolding

5.3 CP-2 Thesis-Faithful Target Profile

CP-2 is the intended DSF-faithful architecture: - structural event tape - shared fast/mid/slow memory state - jointly interpreted horizons - materially active negative-space and temporal order - epoch as explicit external field with governed join semantics - deterministic multi-view ensemble and coherence-field-guided governance

6. Mathematical Foundations

6.1 Structural State

For instrument s at evaluation time t , canonical structural state is represented as:

$$X_{s,t} = (regime, S_{UF}, R_{UF}, stability, D, M, R_{rev}, U^*, C, P, B, guard, hardening)$$

where all members are deterministic functions of the input bar sequence up to time t .

6.2 Policy Keying

Let $K(X_{s,t})$ be the structural cell key constructed from a governed subset of structural members. In current production, the key family includes: - regime - D - M sign - R_rev - U bucket - C - P bucket - B sign - S bucket - R bucket - optional CD bucket - optional phase suffix - optional deterministic multi-view suffix

6.3 Ladder Mapping

Decision provenance maps every evaluated row to exactly one ladder level:

$$L(X_{s,t}) \in \{L0, L1, L2, L3, L4\}$$

with deterministic precedence rules: 1. exact structural match 2. relaxed structural match 3. safe policy fallback 4. degraded or unmapped 5. unavailable or blocked

6.4 Structural Recency

For each structural event family e , define deterministic recency:

$$steps_since_e(s, t) = t - \tau_e(s, t)$$

where $\tau_e(s, t)$ is the most recent timestamp at or before t when event e occurred, subject to null and reset rules defined in the member registry.

6.5 Publication Consistency

A publication bundle P_i is valid only if:

$$publication_id(snapshot_i) = publication_id(quote_i) = publication_id(policy_i)$$

and all required freshness and validation gates pass.

7. Canonical L0-L4 Structural Engine

7.1 Purpose

L0-L4 produce deterministic structural interpretation from market time-series. They are the canonical basis for all governed downstream policy use.

7.2 Inputs

- ordered price bars
- bar counts and timestamps
- any formally admitted non-price inputs defined by the active kernel/profile

7.3 Outputs

- regime classification
- structural field strengths
- stability score
- drawdown state
- decision vector components
- decision guard and hardening markers

7.4 Normative Role

L0-L4 are not optional feature engineering. They are the canonical interpreter of structural geometry. Downstream policy layers may transform, serialize, or bucket outputs, but such transforms must be explicit and auditable.

8. Current Production L5 Code-Truth Profile (CP-0)

8.1 Entry Points

The current production path includes: - admin refresh API - refresh wrapper - quote/profile refresh steps - L5 policy learning - oracle short-cycle gate - runtime sync to Postgres - validation gate - recommendations and portfolio advisor read paths

8.2 Policy Generation

Current L5 supports replay, rowtrace, and hybrid policy generation modes. Decisions are selected from explicit actions using weighted excess-vs-SPY or action-return objectives, subject to action edge, margin, and winrate guards.

8.3 Promotion

Promotion requires non-regression across: - anomaly-accounted excess - outcome-over-index - mapping rate

8.4 Advisor Surfaces

Recommendations and portfolio surfaces consume runtime rows and policy matching results through explicit provenance and fallback contracts. They do not directly expose all internal policy machinery.

9. Strict Function-by-Function Lineage

9.1 Lineage Principle

Every user-visible decision must be traceable from: - canonical structural computation - snapshot materialization - runtime sync/publication - policy matching - advisor response shaping

9.2 Key Transform Boundaries

Critical transforms include: - flattening canonical structural outputs into snapshot rows - deriving PSCF key candidates - row-trace export/import - request-time recomputation vs persisted provenance - runtime publication binding

9.3 Current Loss/Transform Points

Current CP-0 loss/transform points include: - snapshot flattening - candidate-key fallback chain relaxation - request-time provenance recomputation (now mitigated by persisted provenance) - row-trace serialization - absent recency fields in live snapshot rows - absent epoch-state join fields in live snapshot rows

10. Provenance, Audit, and Chain-of-Custody

10.1 Decision Provenance

Persisted decision provenance includes, at minimum: - run id - snapshot digest - policy artifact identity - candidate key chain - matched key - ladder level - decision and reason code - anomaly flags - recency components used - epoch components used - provenance schema version

10.2 Audit Expectations

All activation, publication, and runtime transitions must be auditable, timestamped, and reconstructable.

11. Typed Fallback Ladder

11.1 Definition

The ladder is a semantic reporting and control doctrine, not merely a diagnostic label.

11.2 Current Post-A4.2 State

Following exact-path alignment adoption, current production can materially restore exact structural coverage through the no-ST exact family anchor while keeping selected-cell and decision drift at zero in audited shadow/adoption runs.

11.3 Operational Use

The ladder governs: - quality reporting - activation readiness - strategy-family admissibility - promotion-gate design - proof-plane decomposition

12. Exact-Path Construction

12.1 Current Truth

The current exact anchor is adopted as: **NO_ST_EXACT_FAMILY_KEEP_CD_PHASE_SPIDER_STRICT**

12.2 Operational Consequence

This restores material exact coverage without observed selected-cell or decision drift in CP-0 adoption verification.

13. Financial Strategy Rulebook

13.1 Current Coverage State

Current production coverage is dominated by: - long-side accumulation / hold discipline - abstention / suppression / degraded handling

Minor or blocked families include: - valuation-protected accumulation - survivability/capital-preservation protected accumulation - margin-quality discipline - insider/ownership overlays - recency-sensitive accumulation - epoch-assisted accumulation - pattern/setup overlays - company-news/catalyst overlays - sector/macro/epoch intelligence - short-side/downside families

13.2 Long-Side Decomposition

Current long-side base is candidate context, not automatic action. Subfamilies include: - accumulation_core - hold_discipline_core - hold_only_no_add_discipline

13.3 Strategy Family Gating by Ladder

The system distinguishes families that: - require L0 exact - may operate at L1 relaxed - are safe-only on L2/L3/L4

14. Quote-Family Contract State

14.1 Current Truth

Quote-family contracts are complete, but quote-family activation is blocked by source freshness/publication alignment problems, not schema absence.

14.2 Current Blockers

- publication alignment with snapshot = false in stale-state investigations
- quote age within validation window = false in stale-state investigations
- quote cache materially stale relative to active snapshot prior to remediation lane work

15. Structural Recency Contract State

15.1 Current Truth

Structural recency contract definition is complete. Activation readiness remains blocked only by missing snapshot materialization fields.

15.2 Required Additions

- `steps_since_*` top-level snapshot fields
- `structural_recency_schema_version`

15.3 Readiness Status

- runtime transport path: favorable
- provenance consumer path: favorable
- activation: blocked only by missing materialized fields

16. Epoch Contract State

16.1 Current Truth

Epoch contract definition is complete but activation readiness remains more blocked than recency.

16.2 Current Blockers

- numeric `regime_code` not yet stable
- `gap_state` enum not defined in current code truth
- sidecar projection join not implemented

16.3 Required Additions

- governed regime_code enum contract
- governed gap_state enum contract
- epoch schema version
- runtime sidecar projection/join semantics

17. Publication Architecture Hardening Doctrine

17.1 Atomic Publication Bundle

The system shall move toward a single atomic runtime bundle containing: - snapshot artifact - quote artifact - policy artifact - provenance artifact - validation result - publication status

17.2 Immutable Artifacts, Mutable Pointer

Artifacts should be immutable. Runtime readers should use a single mutable active publication pointer.

17.3 Activation Gate

No publication becomes active unless: - component publication identities align - freshness is within SLA - validation passes

17.4 Resumable Publishers

Heavy publishers, especially quote build, must be chunked, observable, resumable, and marker-writing.

18. Quote Freshness and Publication Alignment Doctrine

18.1 Root-Cause Findings

The stale quote problem is decomposed into: - publication alignment failure - refresh bookkeeping gap - serving-status timestamp source mismatch as a separate issue

18.2 Current State in v2.5/v2.6 Context

Workspace/artifact alignment fixes exist. Live alignment remediation remains a separate lane until verified green in the active runtime environment.

18.3 Safety Rule

Quote-family activation remains blocked until quote publication alignment and freshness are confirmed within the active runtime context.

19. Reader Doctrine

Recommendations and portfolio readers must be publication-aware. They must not silently mix components from mismatched publications.

20. Validation, Verification, and Proof Planes

20.1 Plane 1: Structural Quality

- exact vs relaxed coverage
- fallback distribution
- family-level quality

20.2 Plane 2: Operational Quality

- freshness
- publication alignment
- degraded serve percentage
- audit completeness

20.3 Plane 3: Commercial Competitiveness

- overlap against alternative systems
- return/quality on governed slices
- trust and explanation quality

21. Current Competitiveness Choke Map

The current primary chokers are: 1. quote publication alignment/freshness not yet fully green in live runtime 2. recency not yet materialized into live snapshot rows 3. epoch not yet transported/joined into live runtime rows 4. rulebook breadth still narrow beyond dominant long-side and abstention families 5. proof and competitiveness planes still maturing

22. Controlled Improvement Sequence

The current sequence is: 1. A3 provenance persistence 2. A4 fallback ladder 3. A4.1 exact recovery audit 4. A4.2 exact-path adoption 5. A5.1 rulebook coverage 6. A5.2 long-side decomposition 7. A5.3 subfamily contracts 8. A5.4 quote and recency/epoch contracts 9. A5.5 recency and epoch activation readiness 10. parallel quote freshness/publication alignment remediation

23. Change Control and Release Discipline

No behavior-changing adoption may be treated as complete until: - artifacts exist - bounded verification passes - drift counts are recorded - blockers and dependencies are explicit

24. Conclusion

TFE v2.6 reflects a system that has materially improved determinism, provenance, fallback semantics, and exact-path correctness. The next substantive DSF-aligned value lies in structural recency materialization, then epoch transport, while stale quote/publication alignment remains a separate operational blocker that must be cleared for quote-family competitiveness.

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